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INTERNET PROGRAMMMING

CO1 – A2 CASE STUDY

Case Study Assessment: Web Development Standards Implementation

Question 1: Video Embedding Options

Scenario:
A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The team needs to embed a promotional video on the homepage.

Question Text:
Select the correct HTML tag for video embedding and justify your answer.

Data Snippet 1: Video Embedding Options

- A. <media src="video.mp4"></media>
- B. <video src="video.mp4" controls></video>
- C. <movie src="video.mp4"></movie>
- D. <vid src="video.mp4"></vid>

Solution & Justification:

- **Correct Option: B**
- **Detailed Analysis:** The <video> element is the native, standardized HTML5 tag used to embed video content directly into web applications without relying on third-party plugins or external tools. The inclusion of the controls attribute is essential because it instructs the browser to display the default user playback interface, which includes vital user buttons like play, pause, timeline seeking, and volume adjustment. The

other options (<media>, <movie>, and <vid>) are entirely invalid and do not exist within the official World Wide Web Consortium (W3C) HTML specifications.

Implementation Code Block:

```
html

<!-- Native HTML5 video element with playback controls -->

<video src="video.mp4" controls width="640" height="360">

  <p>Your browser does not support the video tag. Please upgrade your browser.</p>

</video>
```

Question 2: Form Submission Methods

Scenario:
A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. A registration form must securely send user data to the server.

Question Text:
Identify the appropriate HTTP method for form submission and explain why.

Data Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Solution & Justification:

- **Correct Option: B**
- **Detailed Analysis:** The POST method is the industry standard for sending sensitive user data, such as registration credentials and passwords, to a server. Unlike the GET method—which appends data

directly into the plaintext URL query string—the POST method packages the submitted data securely within the body of the HTTP request message. This crucial design distinction ensures that private user passwords do not get leaked into public browser histories, server access logs, web caches, or bookmarked URLs.

Implementation Code Block:

```
html

<!-- Secure user registration form utilizing the POST method -->

<form action="/submit-registration" method="POST">

  <div>

    <label for="username">Username:</label>

    <input type="text" id="username" name="username" required autocomplete="username">

  </div>

  <div>

    <label for="password">Password:</label>

    <input type="password" id="password" name="password" required autocomplete="new-password">

  </div>

  <button type="submit">Register Account</button>

</form>
```

Question 3: Navigation Elements

Scenario:
A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The navigation menu must follow semantic HTML standards.

Question Text:
Choose the proper semantic element for navigation and justify its usage.

Data Snippet 3: Navigation Elements

- A.
- B. <nav>
- C. <section>
- D. <header>

Solution & Justification:

- **Correct Option: B**
- **Detailed Analysis:** The <nav> element is an official HTML5 semantic tag explicitly reserved to wrap a collection of primary navigation links. Utilizing targeted semantic tags over generic structural blocks like significantly improves web accessibility and Search Engine Optimization (SEO). It creates a clear data structure, allowing modern assistive tools (such as screen readers used by visually impaired users) to instantly locate, identify, and jump directly to the site navigation layout.

Implementation Code Block:

html

```
<!-- Semantic HTML5 structural block for primary site links -->
<nav aria-label="Main Website Navigation">
  <ul>
    <li><a href="#home">Home</a></li>
    <li><a href="#features">Features</a></li>
    <li><a href="#pricing">Pricing</a></li>
    <li><a href="#contact">Contact Us</a></li>
  </ul>
</nav>
```

Question 4: XML Syntax Errors Analysis

Scenario:
A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. XML configuration files are being used but contain syntax errors.

Question Text:
Identify the correct XML syntax and explain the error in other options.

Data Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from><from>Admin</from></note>
- D. <note to="User" /> </note>

Solution & Justification:

- **Correct Option: B**
- **Detailed Analysis:** Extensible Markup Language (XML) enforces strict formatting guidelines where all components must be fully closed and nested in an orderly hierarchical tree. Option B follows these structural rules perfectly: the main root tag <note> surrounds the child elements <to> and <from>, and every tag opens and closes correctly.
- **Error Analysis of Alternatives:**
    - **Option A:** The opening root tag <note> is completely missing its closing </note> tag at the very end.
    - **Option C:** The opening tag <to> is mistakenly closed with a mismatched tag </from>, breaking tag symmetry rules.
    - **Option D:** Contains highly malformed syntax because it tries to close a parent element (</note>) that was already terminated inline as a self-closing tag (<note to="User" />).

Implementation Code Block:

xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Well-formed XML structural configuration layout -->
<note>
   <to>User</to>
   <from>Admin</from>
</note>
```

Question 5: Web Communication Protocols

Scenario:
A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The application must communicate over the correct web protocol.

Question Text:
Choose the correct protocol for web communication and justify your choice.

Data Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Solution & Justification:

- **Correct Option: B**
- **Detailed Analysis: HTTP** (Hypertext Transfer Protocol) is the foundational application-layer protocol utilized worldwide to transmit web pages, media assets, and data files between client web browsers and servers. The alternate network options are built for completely distinct network utilities:
    - **FTP** (File Transfer Protocol) is built strictly for moving files between storage systems.

- **SMTP** (Simple Mail Transfer Protocol) is engineered exclusively for routing emails.
- **SNMP** (Simple Network Management Protocol) is used by IT administrators to monitor and manage local network hardware.

Implementation Code Block:

javascript

```
// Example of a client web request executed over the HTTP protocol layer
fetch('https://example.com', {
    method: 'GET',
    headers: {
        'Accept': 'application/json'
    }
})
.then(response => response.json())
.then(data => console.log('HTTP Response Received:', data))
.catch(error => console.error('Network Protocol Error:', error));
```